

How to Analyze the Full Mass Balance in Vanadis 3D v3.1

Guide to log messages and OUTPUT/MASS_BALANCE.TXT

This guide explains the full mass-balance diagnostics in the current Vanadis version: source Q, outflow OUT, inflow IN, the DIV term for spatially varying velocity, removal P(S), Robin/deposition ALPHA, the algebraic Dirichlet reaction DIR, and the numerical residual EPS. The examples use 27-point HEX8 and 9-point Q4 quadrature, the latest DR formulation, and CUDA.

Key rule: M(t) alone is not enough to judge mass conservation. Check the balance equation for every time step and the cumulative balance.

1. The equation that must close

For a step from t_n to t_{n+1} :

$$dM = M_{n+1} - M_n$$

Current sign convention:

$$dM = M_Q - M_{out} + M_{in} + M_{div} - M_P - M_{alpha} - M_{dir} + eps_{num}$$

For divergence-free velocity, $M_{div} = 0$. The current test uses $v = (0, -5, 0)$, so DIV is zero.

2. Columns in OUTPUT/MASS_BALANCE.TXT

Column	Meaning	Sign / interpretation
t	end-of-step time [s]	information
M_n	mass at step start [kg]	state
M_np1	mass at step end [kg]	state
dM	$M_{np1} - M_n$ [kg]	left-hand side
M_Q	source mass in the step [kg]	+ gain
M_out	mass leaving boundary [kg]	- loss
M_in	mass entering boundary [kg]	+ gain
M_div	volume contribution from $div(v)$ [kg]	+ in current equation
M_P	removal by P(S) [kg]	- loss
M_alpha	Robin/deposition [kg]	- loss for positive deposition
M_dir	algebraic Dirichlet reaction [kg]	- if positive; negative adds algebraically
eps_num	step balance residual [kg]	should be very small
Cum_*	cumulative sums [kg]	cumulative balance

DIR note: a negative M_{dir} is not automatically a physical inflow through a wall. It is the algebraic reaction of the full discrete system after imposing an essential Dirichlet condition. It may be negative initially and positive later.

3. Reading the balance in the log

After each time step the log prints the mass change, current-step terms, and cumulative sums. For $t = 100$ s:

```
time=100 mass=92.473031089950751
MASS BALANCE [kg]: dM=92.473031089950751 Q=100.0 OUT=0 IN=0
DIV=0 P=8.3668460939 ALPHA=0.3915526120
DIR=-1.2314295689 EPS=2.269206e-7
```

Manual check:

$$100 - 8.3668460939 - 0.3915526120 - (-1.2314295689) + 0.0000002269 = 92.47303108995 \text{ kg}$$

This reproduces dM , so the source, removal, Robin term, and Dirichlet reaction close consistently for this step.

4. Reading OUTPUT/MASS_BALANCE.TXT

Each row is one time step and can be analyzed directly in Excel, Python, or another tool. The simplest row check is:

$$R_{\text{step}} = dM - (M_Q - M_{\text{out}} + M_{\text{in}} + M_{\text{div}} - M_P - M_{\text{alpha}} - M_{\text{dir}})$$

R_{step} should equal eps_num to file precision. A clear mismatch indicates a diagnostic error, file-version mismatch, or a missing balance term.

For cumulative sums with $M(0) = 0$:

$$M(t) = \text{Cum}_Q - \text{Cum}_{\text{out}} + \text{Cum}_{\text{in}} + \text{Cum}_{\text{div}} - \text{Cum}_P - \text{Cum}_{\text{alpha}} - \text{Cum}_{\text{dir}} + \text{Cum}_{\text{eps}}$$

5. Final example: $t = 2000$ s

Quantity	Value [kg]
M(2000)	411.94623624538661
Cum_Q	2000.0000000000000
Cum_out	0
Cum_in	0
Cum_div	0
Cum_P	696.73017801422975
Cum_alpha	39.352025002521110
Cum_dir	851.97156208144497
Cum_eps	1.3435825110e-6

$$2000 - 696.73017801423 - 39.352025002521 - 851.971562081445 + 0.000001343583 = 411.946236245387 \text{ kg}$$

The cumulative relative residual with respect to total source mass is about $6.7\text{e-}10$, i.e. about $6.7\text{e-}8$ %. This is very good mass-balance closure.

6. What to monitor in practice

M(t): mass currently present in the domain. It need not grow as $Q \cdot t$ because $P(S)$, deposition, Dirichlet conditions, and open-boundary transport can act simultaneously.

eps_num: the main per-step control value. It should be very small compared with typical masses exchanged in that step; in the current test it is of order $1e-6$ kg or smaller.

Cum_eps: shows whether small step errors accumulate. If it grows to a value significant compared with Cum_Q or $M(t)$, investigate.

OUT/IN: zero in the current test because exchange associated with active Dirichlet constraints is represented by DIR. They can be nonzero with natural open boundaries.

DIV: near zero for $\text{div}(\mathbf{v})=0$. With a spatially varying velocity field of nonzero divergence, DIV may be genuinely nonzero and is needed for consistency with the $\mathbf{v} \cdot \text{grad}(\mathbf{S})$ form.

P: integral mass removal due to local $P(S)S$. Time variation is natural because P depends on the concentration field.

ALPHA: removal through the Robin/deposition condition. Positive alpha and concentration usually give positive mass loss.

DIR: especially important with five $S=0$ walls. Near steady state, positive DIR may remove a large fraction of the supplied source mass.

7. Practical procedure for a new run

- 1 Check that all values are finite: no NaN/Inf in M , balance terms, or EPS.
- 2 Compare R_step with eps_num at the beginning, transient maximum, and end.
- 3 Check the cumulative equation and $|Cum_eps| / \max(Cum_Q, |M|)$.
- 4 After code changes, compare $M(t)$, the $P(S)$ range, and Cum_eps with the reference run.
- 5 After boundary-condition changes, see which term carries the exchange: OUT/IN, ALPHA, or DIR.

Practical QA indicator (not a rigid mathematical criterion): relative cumulative residual around $1e-8$ or below is excellent; $1e-8$ to $1e-6$ is usually very good; clear growth above $1e-6$ should trigger checks of sources, boundary conditions, and diagnostic consistency.

8. What to archive with the result

For every validation run, keep the standard log, OUTPUT/MASS_BALANCE.TXT, the configuration file, the source-version identifier, and whether coloring or no-color was used. This allows both the physical result and the integral balance closure to be reproduced.

Current reference: coloring run: $M(2000) = 411.94623624538661$ kg, $Cum_eps = 1.3435825e-6$ kg, and the cumulative relative residual with respect to the 2000 kg source mass is about $6.7e-10$.